

Preliminary Investigation of Knowledge Transfer from Skeleton to IMU for Robust Activity Recognition in Industrial Settings

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Abstract

Skeleton-based models have demonstrated superior performance in human activity recognition (HAR) tasks due to their ability to capture rich spatial and global contextual information. Moreover, skeleton data tend to exhibit greater robustness than IMU data under various perturbations, such as changes in motion speed. However, in many real-world scenarios—such as industrial environments—obtaining skeleton data during inference is often infeasible due to occlusions or camera placement constraints, leaving only IMU data available at test time. To address this limitation, we propose a cross-modal knowledge transfer framework that leverages skeleton data available only during training to guide the IMU model in learning key temporal and structural information. Experimental results demonstrate that our method significantly enhances the performance of IMU-based HAR models under accelerated worker motion scenario.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computing methodologies** → *Knowledge representation and reasoning*.

Keywords

Human Activity Recognition, Cross-modal Learning, Knowledge Distillation, Industrial Environment

ACM Reference Format:

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1 Introduction

Human Activity Recognition (HAR) has received increasing attention in real-world applications, particularly in industrial environments [5–7, 10, 11, 13]. Among commonly used sensing modalities, inertial measurement units (IMUs) are favored for their low cost,

compact size, and ease of deployment. However, IMU-based models often suffer from limited spatial awareness and reduced robustness under varying movement speeds. In contrast, skeleton-based models excel in HAR tasks by capturing rich spatial structures, temporal dynamics, and global contextual information, leading to superior performance and greater robustness—especially under large variations in movement speed or amplitude, or when multiple limbs perform complex, coordinated motions [4, 15].

To illustrate the robustness gap under speed variation in manual work in a warehouse, Table 1 compares the F1 scores of the models based on skeletons and IMUs under normal speed (S1) and accelerated (S4) scenario where skeleton/IMU models were trained on S1 data.¹ While the skeleton-based model maintains relatively stable performance, the IMU-based model shows a notable decline, highlighting its vulnerability to variations in movement speed.

Motivated by the complementary strengths of these two modalities, we hypothesize that incorporating skeleton data during training can improve the robustness of IMU-based models. This is particularly important in practical scenarios where skeleton data is typically unavailable during inference due to occlusions, constrained camera placement, or privacy concerns. Therefore, developing a robust IMU-based HAR model that leverages skeleton guidance during training holds significant practical value.

However, cross-modal knowledge transfer between skeleton and IMU data presents substantial challenges due to the large modality gap in terms of structure, semantics, and representation. These difficulties arise from the inherent differences between the two modalities, particularly regarding the availability of spatial information. Not all knowledge encoded in skeleton data can be effectively transferred to IMU data. For instance, spatial configurations and full-body postures are explicitly represented in skeleton sequences, whereas IMU data—especially when collected from a single location such as the right wrist—lack any form of global spatial context. As a result, directly distilling skeleton-based information into an IMU model using conventional knowledge distillation methods is fundamentally ineffective, since such information is absent in the student modality.

In contrast, we observe that skeleton data are effective in identifying temporally salient moments within a sequence—those frames that are most informative for activity recognition. We hypothesize that such temporal keypoints are equally important for IMU data, even if the spatial structures of the two modalities differ. To exploit this insight, we incorporate temporal attention mechanisms into both the skeleton and IMU models, enabling each to highlight crucial time steps within their respective sequences. We then apply

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¹Models were trained on most S1 data and tested on held-out S1 data from the same users, as well as their corresponding S4 (accelerated scenario) data.

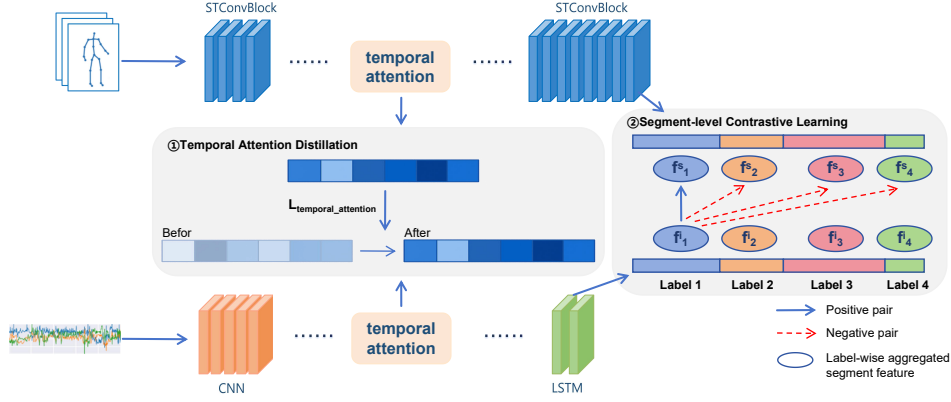


Figure 1: Method Overview

attention-based knowledge distillation to guide the IMU model in mimicking the temporal focus of the teacher model, thereby enhancing its temporal reasoning and improving robustness under varying movement speeds.

Furthermore, in complex manual tasks such as warehouse operations, structural patterns—such as coordinated limb movements and repeated motion cycles—play a critical role in accurate activity recognition. However, such patterns are difficult to transfer through standard distillation. To address this, we propose a segment-level cross-modal contrastive learning approach that leverages the distance between data distributions within the corresponding segments of skeleton and IMU data. Instead of directly mimicking high-level outputs, our method encourages the IMU model to discriminate between structurally matching and non-matching segments. This allows the model to capture high-level structural representations embedded in the skeleton modality, even when such information cannot be explicitly reconstructed from IMU signals.

In summary, our work explores the feasibility and effectiveness of using skeleton data, only available at training time, to improve IMU-based HAR models. We address the modality gap challenge through a carefully designed framework that combines temporal attention distillation and segment-level cross-modal contrastive learning, with the goal of improving recognition robustness under action speed variation.

Table 1: Macro-F1 scores of Skeleton-based and IMU-based models in normal-speed (S1) and accelerated (S4) scenarios.

Model	Modality	S1(%)	S4(%)
ST-GCN	Skeleton	90.28	81.89
Deep-Conv-LSTM	Right wrist IMU	83.54	61.49

2 Related Work

Recent studies have explored cross-modal knowledge transfer for human activity recognition, typically between sensor and vision modalities [3]. However, transferring knowledge specifically from skeleton data to IMU data remains largely unexplored. To our knowledge, only Ni et al. [8] have explicitly investigated this direction,

focusing on the alignment of semantic features. Nevertheless, their method does not account for the complex structural differences between the two modalities, nor does it leverage the skeleton data’s superior ability to identify temporally salient moments within an activity sequence.

In contrast, our work explicitly integrates temporal attention mechanisms and segment-level contrastive learning to address these limitations, thereby significantly enhancing the robustness and generalization of IMU-based models in industrial environments.

3 Methodology

3.1 Overview

We propose a cross-modal learning framework to enhance IMU-based human activity recognition (HAR) by transferring knowledge from a more robust skeleton-based model. As shown in Figure 1, our method combines knowledge distillation [2] and contrastive learning [1] to bridge the modality gap between the skeleton and IMU data.

We adopt ST-GCN[12] as the skeleton-based model and introduce several modifications to further boost its performance. Specifically, we integrate a Convolutional Block Attention Module (CBAM) and temporal attention to improve spatial-temporal feature representation, and introduce residual connections within STConvBlocks to alleviate gradient vanishing issues during training.

For IMU, we use Deep-Conv-LSTM [9] as the IMU-based model, which effectively captures local dependencies through CNN layers and long-term temporal patterns through LSTM.

The knowledge transfer process is made up of two key components: (1) Temporal Attention distillation, and (2) Segment-level Contrastive Learning.

3.2 Temporal Attention Distillation

To transfer the skeleton model’s superior ability to identify temporally salient moments within an activity sequence to the IMU modality, we propose a temporal attention distillation strategy. This module is designed to guide the IMU-based model to mimic the temporal attention distribution of the skeleton-based model, allowing it to focus on the most discriminative time steps within an activity.

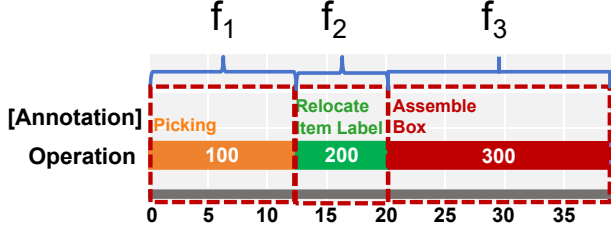


Figure 2: An illustration of label-wise segment aggregation. Features from time steps with the same activity label are averaged to obtain a compact segment representation for contrastive learning.

Specifically, both the skeleton and the IMU models are equipped with temporal attention modules, which output a T-dimensional attention vector indicating the importance of each time step. We extract intermediate temporal attention features from both models and divide them into one-second segments. For each segment, we compute the Mean Squared Error (MSE) between the attention outputs of the skeleton and IMU models. The final temporal attention loss is calculated by averaging the MSE across all segments:

$$L_{\text{temporal_attention}} = \frac{1}{G} \sum_{i=1}^G \text{MSE}(\text{Ske_att}_i, \text{IMU_att}_i) \quad (1)$$

where G denotes the total number of temporal segments, and Ske_att_i and IMU_att_i represent the attention outputs of the skeleton and IMU models at the i -th segment, respectively.

This process enables the IMU-based model to better identify and attend to critical time steps in an activity sequence, thereby enhancing its ability to capture temporal dynamics using only IMU signals.

3.3 Segment-level Contrastive Learning

To effectively transfer the modality-specific structural information from skeleton data to IMU data in complex manual activities, we propose a segment-level contrastive learning strategy. In this approach, we utilize temporal segments annotated with consistent operation labels as comparison units between the skeleton-based and IMU-based models.

Specifically, we first divide both skeleton and IMU data into temporal segments according to operation annotations, where each segment spans from the beginning to the end of a specific operation instance. Within each segment, we extract intermediate features: for the skeleton modality, we use the output of the final ST-ConvBlock of the ST-GCN model, whereas for the IMU modality, we take the output of the first LSTM layer of the Deep-Conv-LSTM model.

To generate a compact and robust segment-level feature representation, we average the extracted frame-level features across all time steps that belong to the same operation label, resulting in a single aggregated feature vector that represents the structural and semantic characteristics of that labeled segment.

As illustrated in Figure 2, these label-wise aggregated segment features serve as inputs for the subsequent contrastive learning stage. Let f_k^s and f_k^i denote the aggregated segment features for the

i -th segment from the skeleton and IMU modalities, respectively. Segment pairs sharing the same operation label form positive pairs, whereas segments associated with different labels constitute negative pairs.

To guide the IMU-based model in learning structural representations aligned with the skeleton modality, we employ the InfoNCE loss:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(f_q^i, f_k^s)/\tau)}{\sum_{k'} \exp(\text{sim}(f_q^i, f_{k'}^s)/\tau)} \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity function, and τ is a temperature hyperparameter. Here, f_q^i represents the anchor segment from the IMU model, while f_k^s and $f_{k'}^s$ represent the corresponding positive and negative segments from the skeleton model, respectively.

The contrastive loss is incorporated into the overall training objective, encouraging the IMU-based model to distinguish structurally aligned segments. Consequently, this method allows the IMU model to capture high-level motion patterns that are inherently challenging to learn from IMU signals alone.

3.4 Loss Function

The overall training objective is defined as a weighted combination of three components: the classification loss, the temporal attention loss, and the segment-level contrastive loss.

The classification loss is the standard cross-entropy loss ($\mathcal{L}_{\text{CE}}$), which encourages the model to correctly predict activity labels from IMU data.

The final objective function is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \cdot \mathcal{L}_{\text{temporal_attention}} + \lambda_2 \cdot \mathcal{L}_{\text{contrast}} \quad (3)$$

where λ_1 and λ_2 are hyperparameters used to balance the three losses.

4 Experiment

4.1 Dataset

We used the OpenPack dataset [14], a large-scale multimodal dataset for the recognition of packing tasks in logistics centers. It contains data from 16 subjects, recorded across four scenarios simulating realistic variations such as procedure changes and product diversity.

S1 represents a standard working scenario where subjects were instructed to follow predefined procedures with limited item variety and no time pressure. In contrast, S4 simulates a busy working environment by introducing periodic auditory alarms that push subjects to speed up their tasks. Additionally, S4 includes irregular operations such as using pre-assembled boxes or grouping small-item orders, making it more complex and dynamic.

To enable effective knowledge transfer between modalities, we preprocessed the raw data to ensure strict temporal alignment. Since the skeleton data was sampled at 15 Hz and the IMU data at 30 Hz, we applied interpolation to upsample the skeleton data to 30 Hz. We then trimmed both modalities to have the same start and end times, resulting in equal sequence lengths and point-to-point correspondence between skeleton and IMU data. This alignment

ensures that each timestamped pair of data can be directly compared during training.

4.2 Experimental Settings

Our goal is to assess the robustness of activity recognition models under varying task speeds. Specifically, we adopt a cross-scenario evaluation setting where all models are trained on S1, which corresponds to normal-speed work scenario, and tested on S4, where workers perform the same tasks at an accelerated scenario. Importantly, both the training and test sets consist of data from the same set of subjects, allowing us to isolate the impact of speed variation while avoiding the confounding effects of cross-user generalization.

Initially, we pretrained the skeleton-based ST-GCN model using S1 data. Training was conducted with a batch size of 16 and a window size of 450, corresponding to approximately 15 seconds of data. An early-stopping strategy was employed to avoid overfitting.

Subsequently, the pretrained ST-GCN model was employed to guide the training of the Deep-Conv-LSTM model via knowledge transfer, using the same S1 dataset. The Deep-Conv-LSTM model utilized identical training configurations as the ST-GCN model, except the batch size was increased to 64. Additionally, we trained a standard Deep-Conv-LSTM model without any knowledge transfer as a baseline for comparative purposes. All model predictions were made at a granularity of one second.

All experiments were conducted on a computational platform equipped with eight Nvidia GTX TITAN X GPUs and a 96-core Intel Xeon Platinum 8163 CPU.

4.3 Result

Table 2 summarizes the model performance evaluated on the S4 scenario of the OpenPack dataset [14]. The baseline refers to the Deep-Conv-LSTM model trained solely on IMU data without any guidance from ST-GCN model. We observe that incorporating Temporal Attention Distillation (172) improves the F1 score from 61.49% to 65.84%, indicating that transferring temporal saliency from the skeleton modality helps the IMU model better identify discriminative moments in an action sequence.

Similarly, applying Segment-level Contrastive Learning (173) improves the F1 score from 61.49% to 63.68%, suggesting that structural information embedded in skeleton segments can guide the IMU model in learning more robust representations.

When combining both techniques (172 + 173), the model achieves the highest F1 score of 66.78%, outperforming the baseline by 5.29%. These results demonstrate the complementary effects of temporal attention guidance and structural alignment in enhancing the robustness of IMU-based recognition under accelerated and variable motion scenario.

5 Discussion

The qualitative analysis in Figure 3 showcases how our proposed method improves IMU-based activity recognition in the accelerated scenario (S4). Two key segments (highlighted in red) illustrate areas where the proposed method significantly outperforms the baseline.

In the first region (around frames 240–310), the baseline model transitions prematurely from class 100 to 200, likely due to its sensitivity to fluctuations introduced by faster movements. In contrast,

Table 2: Macro-F1 scores comparison on S4 scenario of the OpenPack dataset. The number in parentheses indicates improvement over the baseline.

Method	S4 (%)
Baseline	61.49
Baseline + 172	65.84 (+4.35↑)
Baseline + 173	63.68 (+2.19↑)
Baseline + 172 + 173	66.78 (+5.29↑)
Temporal Attention Distillation	
Segment-level Contrastive Learning	

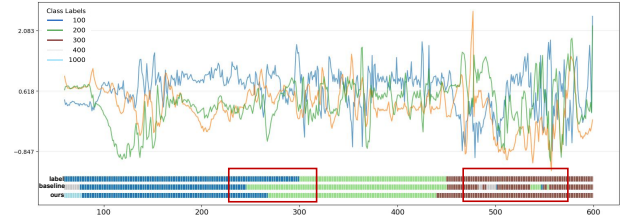


Figure 3: Visualization of IMU Signals and Model Predictions under accelerated scenario (S4)

our method delays this transition and achieves better temporal alignment with the ground truth.

The second region (frames 460–560) reflects another common failure case: fragmented predictions caused by rapid changes in activity structure. Here, the baseline model produces frequent misclassifications, while our method maintains consistent predictions. This improvement results from our segment-level contrastive learning strategy, which helps the model learn more coherent high-level representations across modalities.

Together, these findings highlight the benefit of incorporating skeleton-based temporal and structural supervision during training. This enhancement is particularly valuable in industrial scenarios, where IMU data is often the only reliable source available at test time due to practical constraints such as occlusions or privacy concerns.

6 Conclusion

In this study, we explore the transfer of knowledge from the skeleton to the IMU data through cross-modal means to improve the robustness of activity recognition in industrial environments. By combining temporal attention distillation and segment-level contrastive learning, our method improved the performance of the IMU model under speed variations. Experiments on the OpenPack dataset confirmed its effectiveness.

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References

- [1] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A Simple Framework for Contrastive Learning of Visual Representations. In

- Proceedings of the 37th International Conference on Machine Learning*. 1597–1607.
- [2] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531 [stat.ML] Presented at Deep Learning and Representation Learning Workshop, NIPS 2015.
 - [3] Quan Kong, Ziming Wu, Ziwei Deng, Martin Klinkigt, Bin Tong, and Tomokazu Murakami. 2019. MMAct: A Large-Scale Dataset for Cross Modal Human Action Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 8658–8667.
 - [4] Teerawat Kumrai, Joseph Korpela, Takuya Maekawa, Yen Yu, and Ryota Kanai. 2020. Human activity recognition with deep reinforcement learning using the camera of a mobile robot. In *Proceedings of 2020 IEEE international conference on pervasive computing and communications (PerCom)*. IEEE, 1–10.
 - [5] Takuya Maekawa, Daisuke Nakai, Kazuya Ohara, and Yasuo Namioka. 2016. Toward practical factory activity recognition: unsupervised understanding of repetitive assembly work in a factory. In *UbiComp 2016*. 1088–1099.
 - [6] Jaime Morales, Qingxin Xia, Naoya Yoshimura, Hirotomo Oshima, Masamitsu Fukuda, Yasuo Namioka, and Takuya Maekawa. 2025. Multilevel Transfer Learning for Complex Work Activity Recognition in Logistic Domain. In *2025 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 159–170. <https://doi.org/10.1109/PerCom64205.2025.00036>
 - [7] Jaime Morales, Naoya Yoshimura, Qingxin Xia, Atsushi Wada, Yasuo Namioka, and Takuya Maekawa. 2022. Acceleration-based Human Activity Recognition of Packaging Tasks Using Motif-guided Attention Networks. In *Proceedings of the IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 1–12.
 - [8] Jianyuan Ni, Anne H.H. Ngu, and Yan Yan. 2022. Progressive Cross-modal Knowledge Distillation for Human Action Recognition. In *Proceedings of the 30th ACM International Conference on Multimedia (MM'22)*. 5903–5912.
 - [9] Francisco Javier Ordóñez and Daniel Roggen. 2016. Deep Convolutional and LSTM Recurrent Neural Networks for Multimodal Wearable Activity Recognition. *Sensors* 16, 1 (2016), 115.
 - [10] Qingxin Xia, Joseph Korpela, Yasuo Namioka, and Takuya Maekawa. 2020. Robust Unsupervised Factory Activity Recognition with Body-worn Accelerometer Using Temporal Structure of Multiple Sensor Data Motifs. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 3, Article 97 (Sept. 2020), 30 pages. <https://doi.org/10.1145/3411836>
 - [11] Qingxin Xia, Atsushi Wada, Joseph Korpela, Takuya Maekawa, and Yasuo Namioka. 2019. Unsupervised Factory Activity Recognition with Wearable Sensors Using Process Instruction Information. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 2, Article 60 (June 2019), 23 pages. <https://doi.org/10.1145/3328931>
 - [12] Sijie Yan, Yuanjun Xiong, and Dahua Lin. 2018. Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.
 - [13] Naoya Yoshimura, Takuya Maekawa, Takahiro Hara, Atsushi Wada, and Yasuo Namioka. 2022. Acceleration-based activity recognition of repetitive works with lightweight ordered-work segmentation network. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2 (2022), 1–25.
 - [14] Naoya Yoshimura, Jaime Morales, Takuya Maekawa, and Takahiro Hara. 2024. Openpack: A large-scale dataset for recognizing packaging works in iot-enabled logistic environments. In *2024 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, 90–97.
 - [15] Yiqing Zhang and Takuya Maekawa. 2025. InterHandNet: Capturing Two-hand Interaction for Robust Hand-washing Activity Recognition. In *2025 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 13–24. <https://doi.org/10.1109/PerCom64205.2025.00021>